J. Mol. Biol. (1982) **158**, 567–572

LETTER TO THE EDITOR

Three-dimensional Crystals of a Membrane Protein Complex

The Photosynthetic Reaction Centre from *Rhodopseudomonas viridis*

Reaction centres of photosynthesis isolated from the photosynthetic bacterium *Rhodopseudomonas viridis* by a single step of molecular sieve chromatography were crystallized. The isolated and the crystallized reaction centres contain four different protein subunits, including a membrane-bound cytochrome. Crystallization was achieved by salt precipitation using 2 to 3 M-ammonium sulphate as a precipitating agent in the presence of *N*,*N*-dimethyldodecylamine *N*-oxide as detergent and heptane-1,2,3-triol. Large tetragonal crystals of space group $P4_12_12$ or its enantiomorph diffracting to beyond 2·5 Å were obtained.

High-resolution structural studies on proteins have been important in the elucidation of enzymatic mechanisms and, to date, the structures of more than 50 soluble proteins have been reported. It is important to extend such studies to membrane proteins as processes catalysed by membrane proteins, such as electron and ion transport, light energy conversion and signal transduction, are not well-understood. High-resolution structural analysis requires crystals and only recently have any membrane proteins been crystallized (Michel & Oesterhelt, 1980; Garavito & Rosenbusch, 1980). Here, it is demonstrated that a membrane protein complex, the photosynthetic reaction centre isolated from the bacterium *Rhodopseudomonas viridis*, can be crystallized. The crystals obtained will probably allow a high-resolution structure analysis of this complex, which converts the energy of captured sunlight into electrical and chemical energy in the first steps of photosynthesis.

The successful crystallization of bacteriorhodopsin (Michel & Oesterhelt, 1980), the retinal-dependent light energy converting protein of the halobacteria, led to attempts to crystallize photosynthetic reaction centres. In the photosynthetic membranes, these reaction centres are surrounded by light-harvesting antenna pigment–protein complexes. They catalyse the central primary reaction of photosynthesis. Light, absorbed either by the antenna pigments or the reaction centres drives the transport of electrons across the photosynthetic membranes, thereby creating electric potential.

Among the reaction centres, those isolated from photosynthetic purple bacteria are best-characterized. Normally, they consist of three protein subunits, which are associated with four bacteriochlorophyll, two bacteriopheophytin molecules, one electron-accepting iron–quinone complex and an additional quinone (for a review, see Feher & Okamura, 1979). In several cases, they also contain tightly bound cytochrome molecules, which feed electrons back to the primary electron donor after the light-driven electron transfer.

R. viridis, which contains bacteriochlorophyll b was chosen for crystallization

0022–2836/82/190567–06 $03.00/0

attempts, since in this organism the reaction centres and the light-harvesting antenna pigment–protein complexes form two-dimensionally crystalline membranes (Giesbrecht & Drews, 1966; Wehrli *et al.*, 1979; Miller, 1979). Hexagonal as well as tetragonal lattices are found in the photosynthetic membranes of this organism (Welte *et al.*, 1981). This was taken as proof that the proteins of these photosynthetic reaction centres can make ordered contacts with neighbouring proteins, which is necessary for crystallization in three dimensions. In addition, the photosynthetic reaction centres from *R. viridis* are well-characterized spectroscopically and, to a lesser extent, chemically (e.g. see Thornber & Thornber, 1980; Thornber *et al.*, 1980).

For isolation of the reaction centres a new, simple method was developed. The bacteria were broken up by sonication and crude photosynthetic membranes were isolated by differential centrifugation (100,000 ***g*** for 1 h) after removing debris (22,000 ***g***, 20 min). They were further purified by isopycnic centrifugation on sucrose density-gradients (25% to 50% (w/w), 14 h, 100,000 ***g***). The photosynthetic membranes (broad band at about 35% sucrose) were collected, washed free from sucrose and used for isolation of the reaction centres. Pure reaction centres were obtained in a single step by molecular sieve chromatography (see Fig. 1) after solubilization with *N*,*N*-dodecyldimethylamine *N*-oxide as detergent.

The spectra of the isolated reaction centres correspond to those published (Thornber & Thornber, 1980; Thornber *et al.*, 1980) with an absorbance ratio (280 nm/830 nm) of 2·1 for the purest preparations. However, the reaction centres have probably lost some of the electron-accepting quinones, since substantial

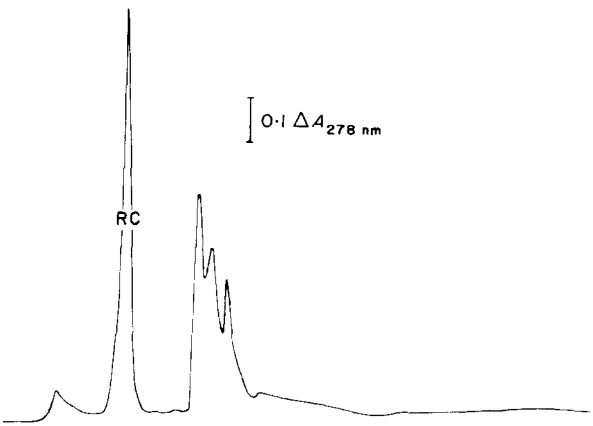

FIG. 1. Elution profile of the molecular sieve chromatography step used for the isolation of the photosynthetic reaction centres from *R. viridis*. Photosynthetic membranes (16 mg of protein/ml) were solubilized in 5% (w/v) *N*,*N*-dodecyldimethylamine *N*-oxide, 0·1% (w/v) NaN_3, 10 mM-Tris·HCl (pH 7) at room temperature for 5 min. After a clarifying spin (100,000 ***g***, 30 min), 0·25 ml of the supernatant was applied to a TSK 3000 SW column (LKB, Gräfelfing) and run at 0·1 ml/min in 20 mM-sodium phosphate (pH 7), 0·1% (w/v) *N*,*N*-dodecyldimethylamine *N*-oxide, 0·1% (w/v) NaN_3. The reaction centres form the peak marked RC.

Due to increasing absorption effects, the detergent concentration had to be raised to 0·5% (w/v) after several runs.

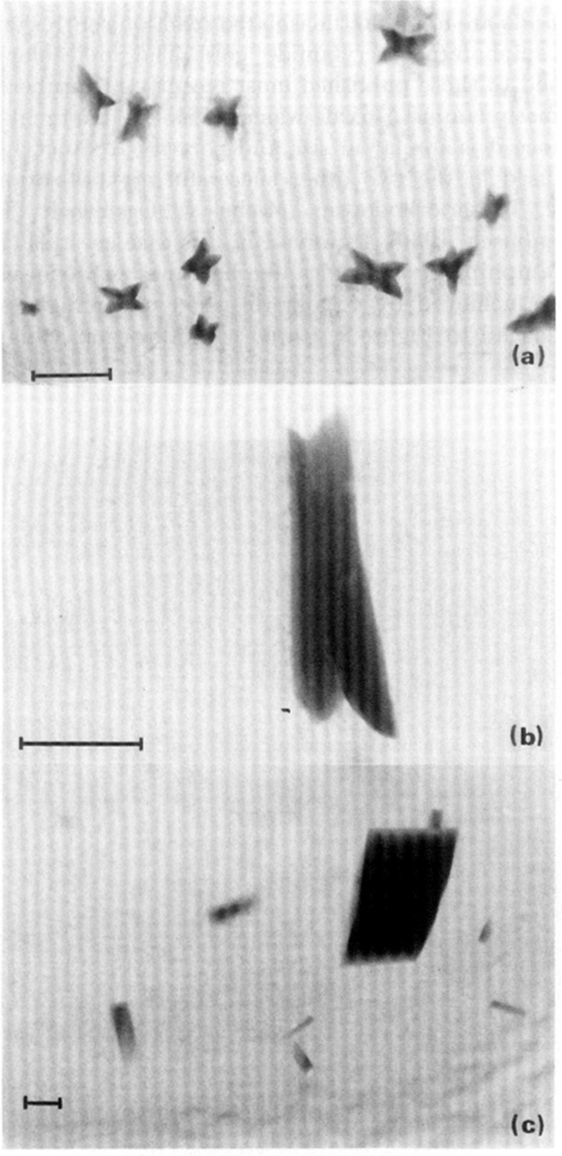

FIG. 2. Crystals obtained by vapour diffusion of a concentrated reaction centre solution (2 units of absorbance at 830 nm) in 1·5 M-ammonium sulphate against: (a) 3 M, (b) 2·6 M and (c) 2·4 M ammonium sulphate at room temperature, pH 6. The fractions containing the reaction centres (see Fig. 1) were pooled and concentrated 6-fold in a desiccator over P_2O_5. Then 3% heptane-1,2,3-triol, 3% (w/v) triethylammonium phosphate and 1·5 M-ammonium sulphate (final concentrations) were added and the pH adjusted. Before crystallization, some denatured protein was removed by centrifugation (2 min, 8000 g). The bar represents 0·1 mm.

bleaching of the primary electron donor P 960, probably a chlorophyll dimer absorbing light of 960 nm wavelength, occurred only after addition of ubiquinone. Four protein subunits are found in the isolated reaction centres when examined on sodium dodecyl sulphate/polyacrylamide gels. The molecular weights of the subunits correspond to those published most recently (Thornber *et al.*, 1980).

N,*N*-dodecyldimethylamine *N*-oxide was used as the detergent for isolation and crystallization, since it is similar in size and polarity to β-octyl-glucopyranoside, which has been used in the two other successful crystallizations of membrane proteins: namely, bacteriorhodopsin (Michel & Oesterhelt, 1980) and porin (Garavito & Rosenbusch, 1980). However, in this case crystals could be obtained only when small amphiphilic molecules like heptane-1,2,3-triol were also present. Crystallization was achieved by vapour diffusion under high salt conditions as described in the legend to Figure 2. Rapid crystallization leads to small, yellow

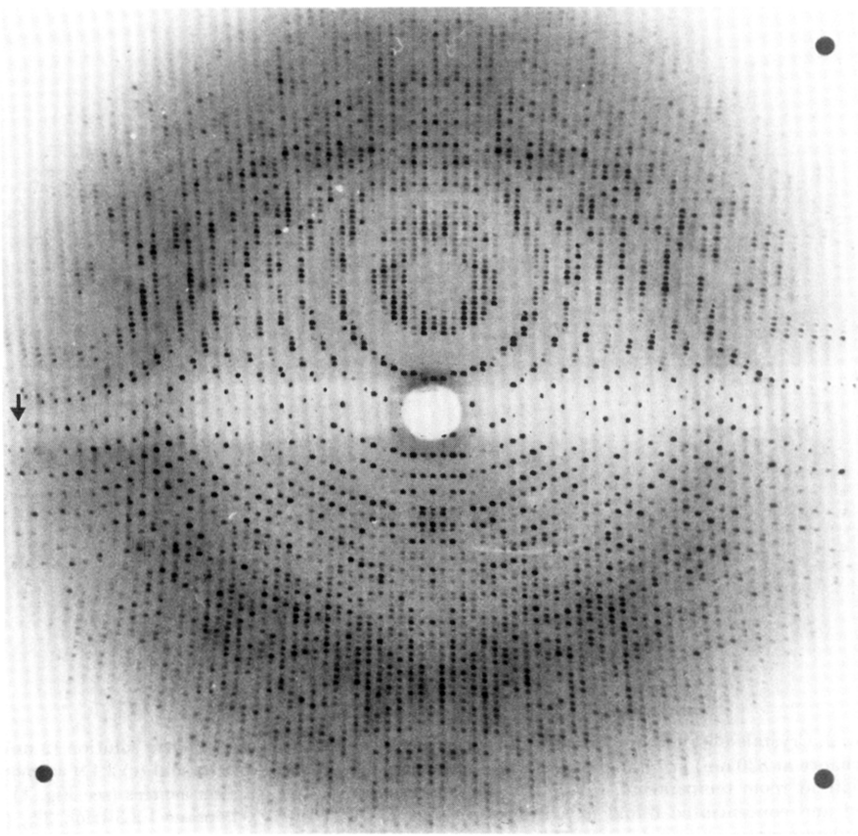

FIG. 3. X-ray diffraction pattern of a single reaction centre crystal (rotation photograph, 1° of rotation around the *c*-axis). The exposure time was 20 h using monochromatized X-rays from a Rotaflex rotating anode (Rigaku, Japan) operated at 45 kV and 90 mA, and a 0·4 mm collimator. The focus to crystal distance was 30 cm. The arrow indicates 3·0 Å resolution.

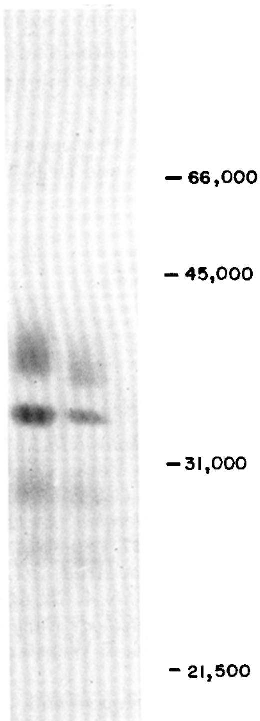

FIG. 4. Comparison of the protein composition of the isolated reaction centre and the crystals on a sodium dodecyl sulphate/10% (w/v) polyacrylamide gel according to Laemmli (1970). Isolated reaction centres (left lane) and a single resolubilized crystal (right lane) were applied in sample buffer without heating to avoid aggregation of the 2 smaller subunits (Thornber *et al.*, 1980). The position of molecular weight markers is given on the right.

starlike crystals (Fig. 2(a)), whereas upon slow crystallization large, dark brown square columns are obtained (Fig. 2(c)). Starting with a protein concentration of 5 mg/ml, they often grow to a size of 0·4 mm × 0·4 mm × 1 mm within three weeks. This size is sufficient to locate the orientations of the different chromophores relative to the crystal axes by various kinds of spectroscopy.

The best crystals diffract X-rays to beyond 2·5 Å resolution. The space group of these tetragonal crystals is $P4_12_12$ or $P4_32_12$, as indicated by symmetries and systematic absences in the X-ray diffraction patterns with unit cell dimensions of 223 Å × 223 Å × 114 Å. Figure 3 shows a rotation photograph as used for data collection.

Figure 4 demonstrates that all four different protein subunits of the isolated reaction centres are also present in the crystals, including the membrane-bound cytochrome (uppermost band of $M_r \sim 40{,}000$ in Fig. 4; see Thornber *et al.*, 1980). Despite the large asymmetric unit, probably containing two molecules of the reaction centre, the crystals will hopefully lead to a high-resolution structure analysis of this fundamental complex, revealing the structure of the reaction centre proteins, the arrangement of the different pigments and the structure of a membrane-bound cytochrome. The mechanism and the reaction sequence of the primary charge separation in photosynthesis might then be understood.

I thank Professor D. Oesterhelt for support, Professor R. Huber and his colleagues for help in the X-ray experiments, Professor H. Scheer and Dr E. Wehrli for material help, and Dr K. Williams for reading the manuscript.

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Received 23 February 1982

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Edited by J. C. Kendrew